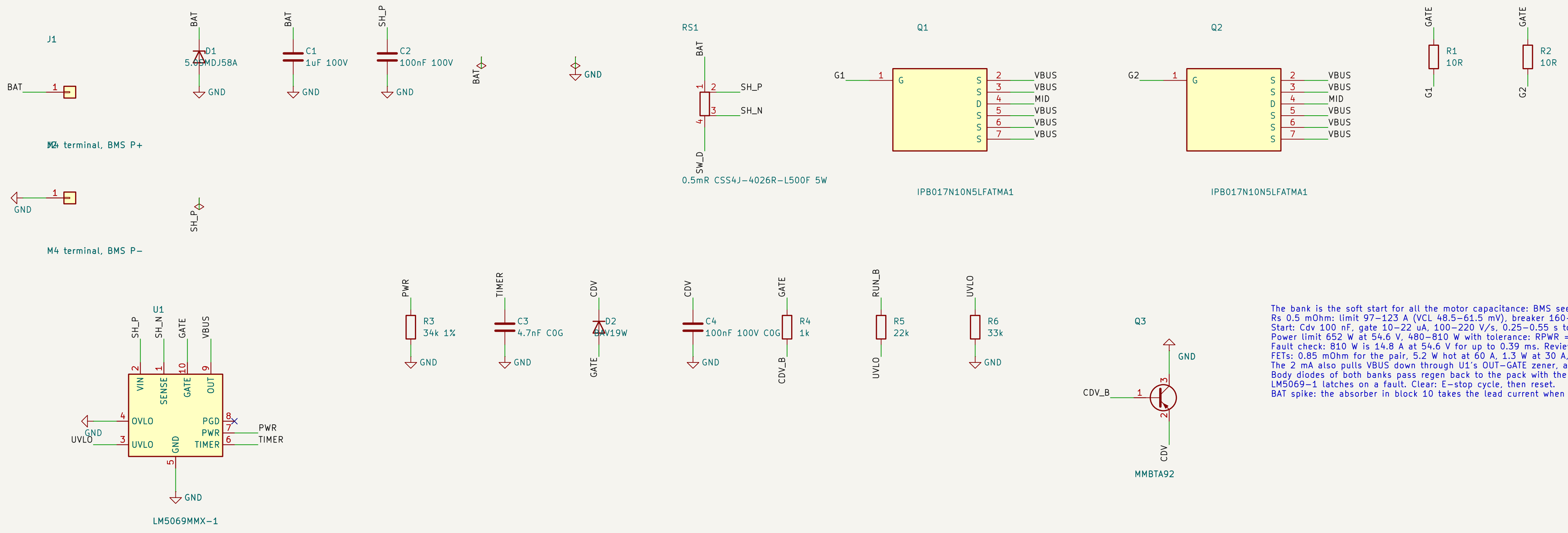
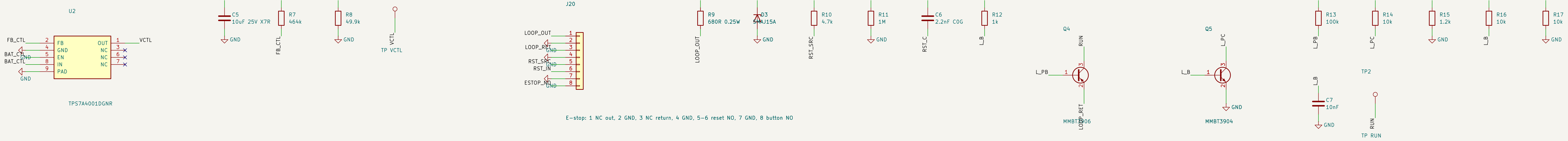
1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7


J1
BAT
M6 terminal, BMS P+
GND
M4 terminal, BMS P-
D1 5.0SMDJ58A
C1 1uF 100V
C2 100nF 100V
RS1 0.5mR CSS4J-4026K-L500F 5W
SH_P
SH_N
Q1 IPB017N10N5LFATMA1
Q2 IPB017N10N5LFATMA1
VBUS
MID
R1 10R
R2 10R
U1 LM5069MMX-1
R3 34k 1%
C3 4.7nF C0G
D2 15V
C4 100nF 100V C0G
R4 1k
R5 22k
R6 33k
Q3 MMBTA92
CDV_B

The bank is the soft start for all the motor capacitance: BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it.
Rs 0.5 mOhm: limit 97–123 A (VCL 48.5–61.5 mV), breaker 160–260 A (VCB 80–130 mV). Above peak draw and JK DOC 80 A. 1.8 W at 60 A.
Start: Cdv 100 nF, gate 10–22 uA, 100–220 V/s, 0.25–0.55 s to full. Bus ceiling 9.4 mF at 2.06 A hot DC SOA line (112 W, 45 K Tj rise). Drive caps unmeasured.
Power limit 652 W at 54.6 V, 480–810 W with tolerance: RPWR = 1.3e5 x Rs x (P – 1.18 mV x VDS / Rs), SNVS452G eq 10, VSNS 6 mV over the 5 mV floor. Timer 4.7 nF: 0.22 ms typ, 0.15–0.39 ms.
Fault check: 810 W is 14.8 A at 54.6 V for up to 0.39 ms. Reviewers read about 16 A at 0.82 ms, 70 C, one FET; TI scales as t^0.346, so about 20.7 A at 0.39 ms: 1.4 x. Read the real curve at layout.
FETs: 0.85 mOhm for the pair, 5.2 W hot at 60 A, 1.3 W at 30 A, the same again in bank A. UVLO low sinks the gate at 2 mA: about 170 us off.
The 2 mA also pulls VBUS down through U1's OUT-GATE zener, against the 54 uA of its internal 1 MOhm SENSE-OUT path, so VBUS reads near 0 V with the banks open.
Body diodes of both banks pass regen back to the pack with the banks open. OVLO tied low: opening on overvoltage would strand regen on the bus.
LM5069-1 latches on a fault. Clear: E-stop cycle, then reset.
BAT spike: the absorber in block 10 takes the lead current when a bank opens fast (breaker trip 1.2 us), 5.0SMDJ58A is the backstop. FETs and U1 are 100 V.

2. Control rail and E-stop latch (no logic IC)

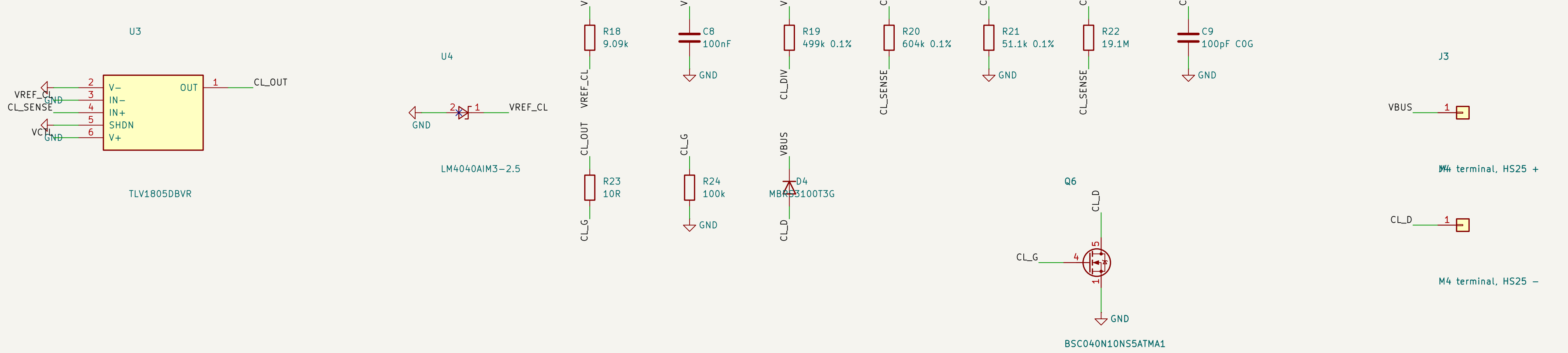

U2 TPS7A4001DGNR
FB_CTL
BAT_CTL
VCTL
C5 10uF 25V X7R
R7 464k
R8 49.9k
TP1 TP VCTL
J20
LOOP_OUT
LOOP_RET
RST_SRC
RST_IN
ESTOP_NO
R9 680R 0.25W
D3 SMAJ15A
R10 4.7k
R11 1M
C6 2.2nF C0G
R12 1k
Q4 MMBT3906
Q5 MMBT3904
R13 100k
R14 10k
R15 1.2k
R16 10k
R17 10k
C7 10nF
TP2 TP RUN

E-stop: 1 NC out, 2 GND, 3 NC return, 4 GND, 5–6 reset NO, 7 GND, 8 button NO

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V.
Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

Seal-in latch, ngspice checked at hFE 80: stays off at power-up even with reset held, reset closes it, a 50 us contact bounce rides through,
loop open drops it in about 100 us, re-closing the loop does nothing, a reset held through a loop cycle does nothing, a new press works 5 ms after release.
RUN 7 V. UVLO and EN_A each take RUN through their own diode (block 10), so a fault on either pin cannot drive RUN. UVLO 3.8 V; needs about 4.8 V on RUN_B rising. Loses VCTL, RUN drops, UVLO drops: bank opens.
Loop 6 mA, wetting for a control-rated NC contact. No 12 V on the connector without a series R. GND between loop out and return, and between return and reset: a short there opens the loop or does nothing.
Brake resistor NC thermostat: wire it in series in the loop harness, no board parts.
Button HB2-BS545, 22 mm mushroom, twist release, 1NC + 1NO, 660 V AC 10 A, control-rated only: NC in this loop, NO to the Pi (block 9). RUN also enables bank A (block 7) and SPARE_SW (block 8).

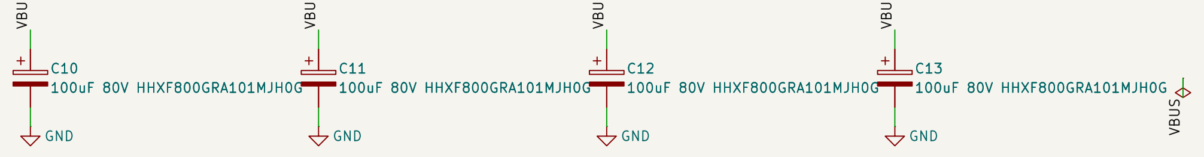
3. Braking clamp: 56.8 V on, 55.8 V off, 57.5 V worst case, into the HS25


U3 TLV1805DBVR
VREF_CL
CL_SENSE
CL_OUT
U4 LM4040AIM3-2.5
R18 9.09k
C8 100nF
R19 499k 0.1%
R20 604k 0.1%
R21 51.1k 0.1%
R22 19.1M
C9 100pF C0G
R23 10R
R24 100k
D4 MBR0100T3G
Q6 BSC040N10NS5ATMA1
CL_G
J3
VBUS
M4 terminal, HS25 +
CL_D
M4 terminal, HS25 –

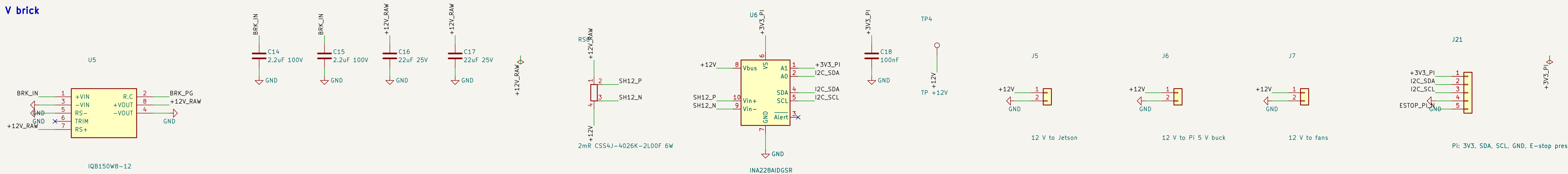
Arcol HS25 10 Ohm on the frame plate: 322 W, 5.7 A. Threshold values are rev A's, unchanged.
Self-powered: VCTL lives as long as the bus or the pack does. If the FET sticks on, the plate thermostat opens the loop and the bank drops the bus.
TLV1805 drives the FET directly: 100 mA peak, 0.3 us into 4 nF. Rev A's driver, meter, shunt, status FET and NTC are gone, and a smaller FET replaces the IPT015.
Leg stage later: fit 2 Ohm and recheck this FET, 29 A gives about 5 W.

4. Bus capacitors

400 uF, 80 V. The drives have their own. Can is 10 x 16.5: land owed, and it only clears under the higher ceiling, away from the fans.

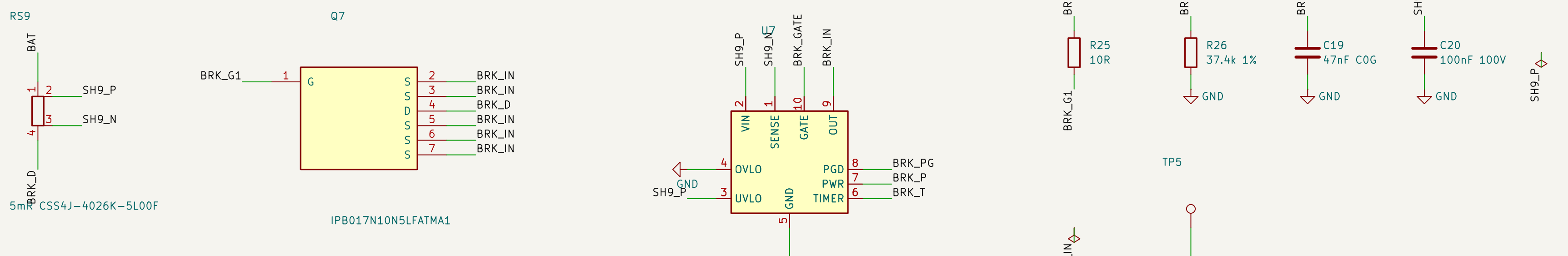

VBUS
C10 100uF 80V HHXF800GRA101MJH0G
C11 100uF 80V HHXF800GRA101MJH0G
C12 100uF 80V HHXF800GRA101MJH0G
C13 100uF 80V HHXF800GRA101MJH0G
GND

5. 12 V brick


U5 IQB150W8-12
BRK_IN
BRK_PG
+12V_RAW
+VIN
-VIN
RS-
TRIM
RS+
R.C
+VOUT
-VOUT
C14 2.2uF 100V
C15 2.2uF 100V
C16 22uF 25V
C17 22uF 25V
RS8 2mR CSS4J-4026K-2L00F 6W
SH12_P
SH12_N
U6 INA228AIDGSR
C18 100nF
TP4 TP +12V
J5 12 V to Jetson
J6 12 V to Pi 5 V buck
J7 12 V to fans
J21 Pi: 3V3, SDA, SCL, GND, E-stop pressed (low)

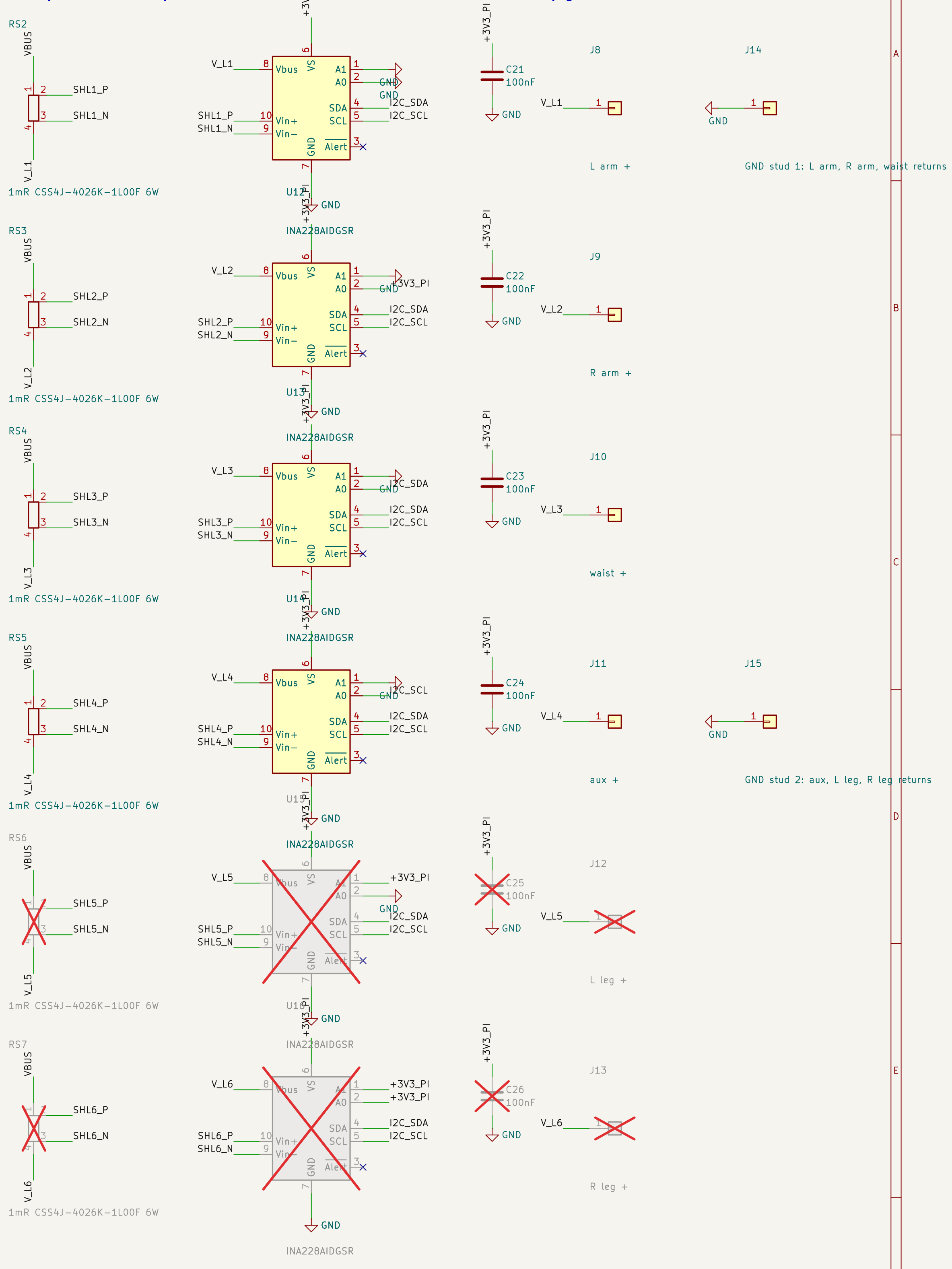
Ahead of the E-stop: motors off, compute stays up. Soft start in block 5a, BRK_PG on R.C keeps the brick off until its input is up.
–Vout is tied to GND here: the robot has one ground, P–. RS+ and RS– tied at the pins, Trim open. Input caps at the pins; the EMC filter is left out.
150 W at 12 V, 89 %: 18 W at full load, about 6 W at 50 W. Derating starts near 30 C at 20 LFM: fit a heatsink on the baseplate, in the Z 67 strip, see REV_B_PARTS. 12.7 mm tall: keep clear of the fan gap.
INA228 addresses: L arm 0x40, R arm 0x41, waist 0x42, aux 0x43, L leg 0x44, R leg 0x45, 12 V 0x46, SPARE_SW 0x47, MID 0x48. Powered by the Pi's 3.3 V; Pi pull-ups on I2C.

5a. Brick soft start (ahead of the E-stop switch)


RS9 5mR CSS4J-4026K-5L00F
SH9_P
SH9_N
Q7 IPB017N10N5LFATMA1
BRK_G1
BRK_IN
BRK_D
BRK_P
BRK_T
U7 LM5069MMX-2
R25 10R
R26 37.4k 1%
C19 47nF C0G
C20 100nF 100V
TP5 TP BRK_IN

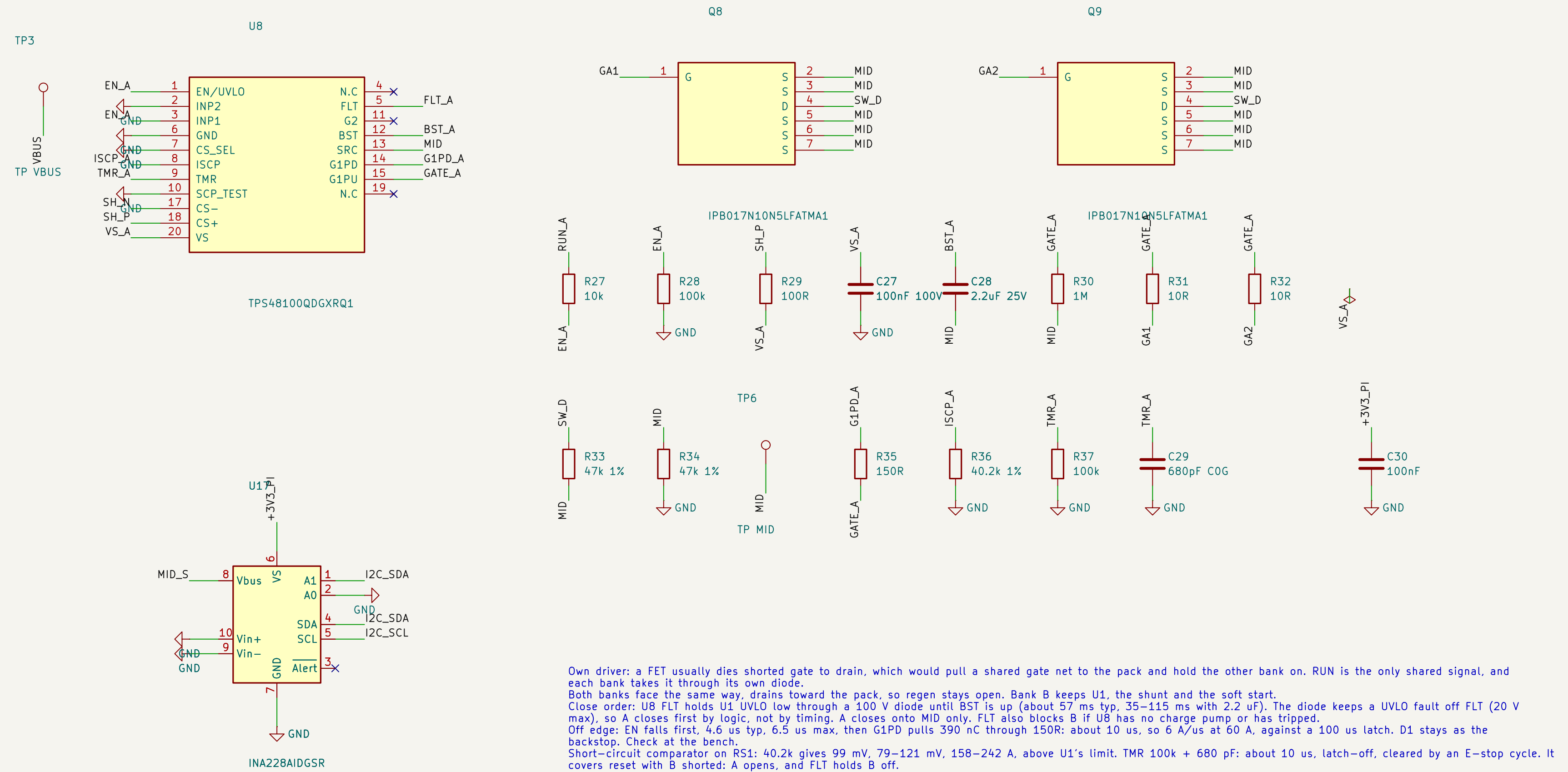
Same controller as the bank, so one part family. –2 retries on its own: a compute fault does not need a BMS cycle. UVLO on VIN, OVLO low: on whenever the pack is.
Power-limit start, no dV/dt network: 70 W limit (Rs 5 mOhm, same equation as U1), about 1.3 A at full pack. 4.4 uF plus the brick's own input caps (not published) charges in under 1 ms at 20 uF.
Timer 47 nF: 2.2 ms typ, 1.5–3.8 ms. Limit 9.7–12.3 A clears the worst brick draw, 4.7 A at 36 V. Brick input capacitance is unpublished: about 40 uF is the most the minimum timer covers, so measure one brick.
Fault check: 87 W is 1.6 A for up to 3.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the linear FET for that reason.
Compute is on whenever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

6. Limb outputs: terminal pair, 1 mOhm shunt, INA228. Fuses inline in the pigtails.


RS2 1mR CSS4J-4026K-1L00F 6W
RS3 1mR CSS4J-4026K-1L00F 6W
RS4 1mR CSS4J-4026K-1L00F 6W
RS5 1mR CSS4J-4026K-1L00F 6W
RS6 1mR CSS4J-4026K-1L00F 6W
RS7 1mR CSS4J-4026K-1L00F 6W
U12 INA228AIDGSR
U13 INA228AIDGSR
U14 INA228AIDGSR
U15 INA228AIDGSR
U16 INA228AIDGSR
INA228AIDGSR
C21 100nF
C22 100nF
C23 100nF
C24 100nF
C25 100nF
C26 100nF
J8 L arm +
J9 R arm +
J10 waist +
J11 aux +
J12 L leg +
J13 R leg +
J14 GND stud 1: L arm, R arm, waist returns
J15 GND stud 2: aux, L leg, R leg returns

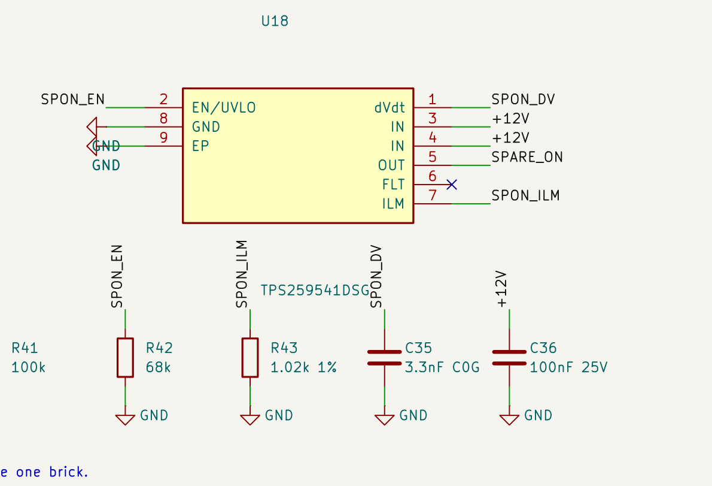
1 mOhm: 60 A peak is 60 mV, ADCRANGE 0. 0.36 W at 19 A.
Legs DNP: fit the same parts when they arrive.
Returns share two GND studs, three limbs each: check the stud and its lug for the summed current.

7. E-stop bank A: second series bank, own driver, ahead of bank B


U8 TPS48100QDGXRQ1
TP3 TP VBUS
Q8 IPB017N10N5LFATMA1
Q9 IPB017N10N5LFATMA1
GA1
GA2
MID
SW_D
R27 10k
R28 100k
R29 100R
C27 100nF 100V
C28 2.2uF 25V
R30 1M
R31 10R
R32 10R
R33 47k 1%
R34 47k 1%
TP6 TP MID
R35 150R
R36 40.2k 1%
R37 100k
C29 680pF C0G
C30 100nF
U17 INA228AIDGSR

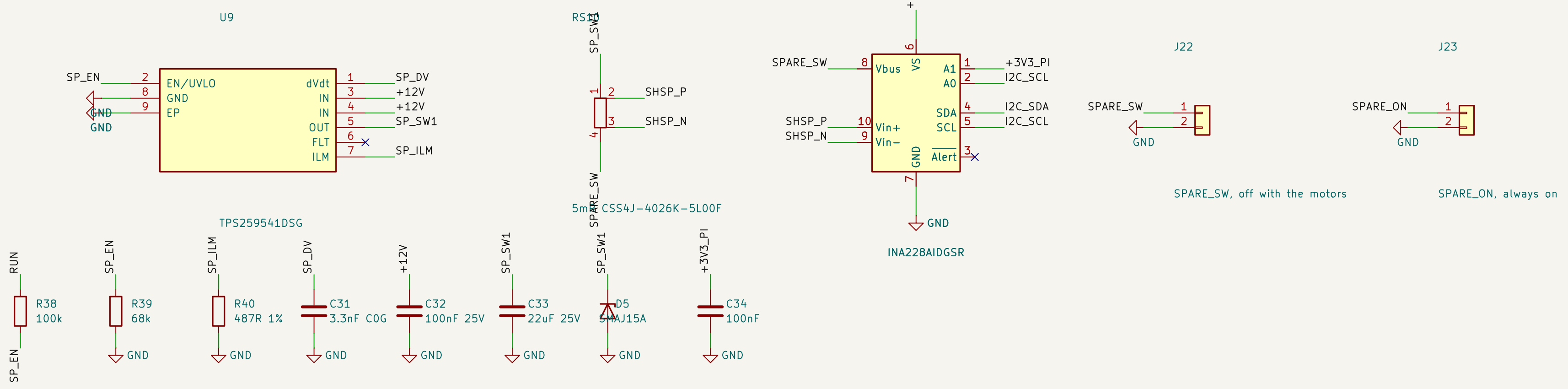
Own driver: a FET usually dies shorted gate to drain, which would pull a shared gate net to the pack and hold the other bank on. RUN is the only shared signal, and each bank takes it through its own diode.
Both banks face the same way, drains toward the pack, so regen stays open. Bank B keeps U1, the shunt and the soft start.
Close order: U8 FLT holds U1 UVLO low through a 100 V diode until BST is up (about 57 ms typ, 35–115 ms with 2.2 uF). The diode keeps a UVLO fault off FLT (20 V max), so A closes first by logic, not by timing. A closes onto MID only. FLT also blocks B if U8 has no charge pump or has tripped.
Off edge: EN falls first, 4.6 us typ, 6.5 us max, then G1PD pulls 390 nC through 150R: about 10 us, so 6 A/us at 60 A, against a 100 us latch. D1 stays as the backstop. Check at the bench.
Short-circuit comparator on RS1: 40.2k gives 99 mV, 79–121 mV, 158–242 A, above U1's limit. TMR 100k + 680 pF: about 10 us, latch-off, cleared by an E-stop cycle. It covers reset with B shorted: A opens, and FLT holds B off.
Driver off holds gate to source with no path to GND, so MID can be biased: 47k + 47k, 0.48–0.50 x pack, 0.6 mA, 63 mW each. Hot FET leakage, up to 200 uA per bank, moves MID by under 5 V. U17 reads MID through 33k (spike and fault limit): calibrate the 3 % gain.
Self-test before reset, pack 36–54.6 V, BMS telemetry for pack voltage: MID 0.4–0.6 x pack and U11 VBUS under 3 V is healthy. MID within 2 V of pack: A shorted. MID under 3 V: B shorted. VBUS high: both shorted, or the bus has not bled down yet, about 11 s on board caps, longer with drives.
U17 uses VBUS only, IN+ and IN– on GND. VBUS alone cannot see one shorted bank, which is why MID is read.

8a. SPARE_ON eFuse


U18 TPS25954DSG
SPON_EN
SPON_DV
SPARE_ON
SPON_ILM
R41 100k
R42 68k
R43 1.02k 1%
C35 3.3nF C0G
C36 100nF 25V

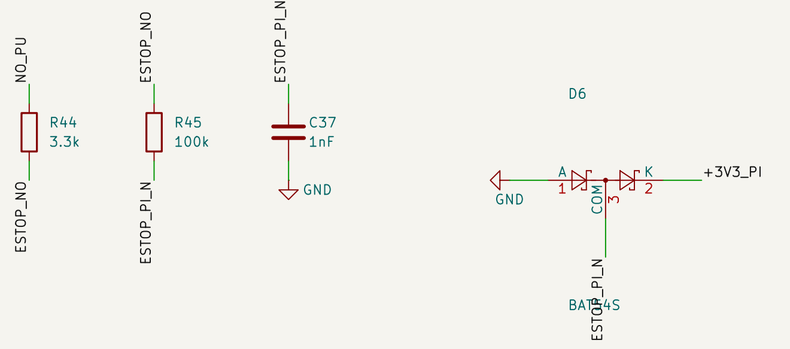
Unswitched, about 2 A limit. Same part as U9: a short stops in 5 us and leaves compute up.

8. Spare 12 V outputs


U9 TPS25954DSG
SP_EN
SP_SW1
SP_ILM
RS10 5mR CSS4J-4026K-5L00F
SHSP_P
SHSP_N
U10 INA228AIDGSR
R38 100k
R39 68k
R40 487R 1%
C31 3.3nF C0G
C32 100nF 25V
C33 22uF 25V
D5 SMAJ15A
C34 100nF
J22 SPARE_SW, off with the motors
J23 SPARE_ON, always on

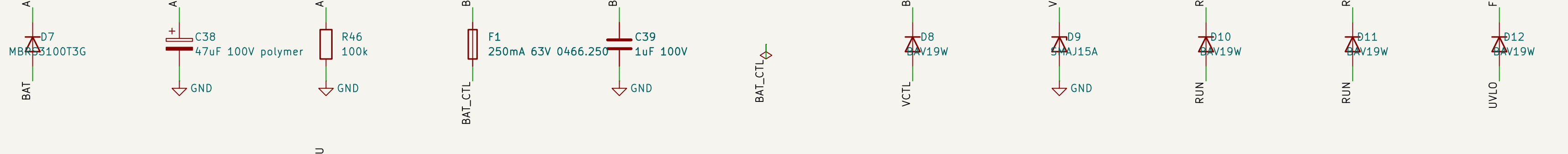
SPARE_SW: RUN enables it, so it drops with the motor bank. The divider keeps 7.7 V of RUN under the 6 V EN limit. Limit 3.87–4.42 A (487R). dV/dt 3.3 nF: about 12 V/ms. One switch: a shorted U9 stays on. Current and volts at 0x47, 5 mOhm, ADCRANGE 1.
SPARE_ON: block 8a. Both spares sit behind RS8, so the 12 V meter counts them.

9. E-stop pressed: button NO contact to the Pi


NO_PU
ESTOP_NO
ESTOP_PI_N
R44 3.3k
R45 100k
C37 1nF
D6 BAT54S
+3V3_PI
GND

Low means pressed. Pi GPIO input, no Pi pull-up needed. 3.3 V through 3.3k gives the contact 1 mA of wetting.
A harness short to the 54.6 V bus: the BAT46W blocks the pull-up (block 10), and 100k leaves 0.5 mA in the clamp pair. Open contact reads about 3.0 V.
Information only: the NC loop does the stopping.

10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)


D7 MBR0100T3G
C38 47uF 100V polymer
R46 100k
F1 250mA 63V 0466.250
C39 1uF 100V
BAT_CTL
D8 BAV19W
D9 SMAJ15A
D10 BAV19W
D11 BAV19W
D12 BAV19W
R47 33k
C40 22nF 100V
D13 BAT46W
C41 330uF 25V

Absorber (B1): a fast breaker trip leaves the pack lead's current with nowhere to go. The diode lets BAT dump it into 47 uF, so a 200 A, 1 uH lead (20 mJ) charges it about 7 V, plus 6 V across a 30 mOhm ESR at 200 A: about 70 V peak. Polymer, ESR 30 mOhm or less. Measure the lead and scope a 60 A stop. 100k bleeds it. The cap charges through the diode when the BMS closes: 47 uF, 0.07 J, check the BMS takes it.
F1, 1 uF, BAV19W and SMAJ15A (M14, M6, M7): a shorted U2 puts the pack on VCTL; the TVS clamps under the 40 V parts and F1 opens. The BAV19W keeps C5 from back-driving U2 if BAT collapses.
RUN_A, RUN_B and FLT_A diodes (B3): a U1 pin 2–3 bridge or an EN fault can no longer drive RUN, so one fault loses one bank at most.
33k on U17 VBUS (M13). 22 nF SRC to GND, TI's recommendation for U8. BAT46W (M4). 330 uF on +12V: spare-output shorts and servo back-feed (M9) land in it and the compute loads.

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR
marengo
Sheet: /
File: marengo_pdu_revb.kicad_sch
Title: marengo power board rev B
Size: A1
Date: 2026-11-06
Rev: B-wip
KiCad E.D.A. 10.0.8
Id: 1/1